# Practice Problem 1

Consider a lawn mower having a sound intensity level of 95 dB at a distance of 2m. Assuming an attenuation coefficient of $10^{-3}\ m^{-1}$ and the sound originates from a point source, determine the sound intensity level at a distance of 200m.

Solution.

$\beta_1 = 95\,dB$ at $r_1 = 2m$

$\beta_2 = ?$ at $r_2 = 200m$

$\alpha = 10^{-3}\,m^{-1}$

point source $\longrightarrow$ $n = 3$

$$\Delta\beta = 10 \log_{10}\left[\left(\frac{r_1}{r_2}\right)^{n-1} e^{-\alpha\Delta r}\right] dB$$

$$= 10 \log_{10}\left[\left(\frac{2}{200}\right)^{2} e^{-10^{-3}(198)}\right]$$

$$= -40.86\,dB$$

$$\beta_2 = (95 - 40 - 78)\,dB$$

$$= 54.14\,dB$$

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# Practice Problem 2

Consider an ultrasonic wave (frequency > 20kHz) generated from a point source. This wave propagates in all directions through water. The sound level is 100 dB at a distance of 4m and 80 dB at a distance of 6m. Determine the attenuation coefficient.

Solution:

$\beta_1 = 100\,dB \qquad r_1 = 4m$

$\beta_2 = 80\,dB \qquad r_2 = 6m$

point source, $n = 3$ } $\Delta\beta = -20\,dB$

$$\Delta\beta = 10 \log_{10}\left[\left(\frac{r_1}{r_2}\right)^{n-1} e^{-\alpha\Delta r}\right]^{1-n}$$

$$\alpha = -\ln\left[10^{0.1\Delta\beta}\left(\frac{r_1}{r_2}\right)\right] \Big/ \Delta r$$

$$\alpha \approx 1.89 \quad m^{-1}$$

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# Practice Problem 3

Consider an object (with mass, m) subjected to 2 driving forces. These forces act separately to produce an oscillation of:

$$y_1(t) = 8\,m \sin\left(\omega t + \frac{\pi}{4}\right)$$

$$y_2(t) = 6m \sin\left(\omega t - \frac{\pi}{6}\right)$$

where $\omega = 127$ rad/s

a) Express these oscillations in phasor form & sketch the phasor amplitudes as vectors.

Solution:

1, Express oscillations in phasor form:

$$y_1(t) = 8 \sin\left(\omega t + \frac{\pi}{4}\right)$$

$$= 8 \cos\left(\omega t + \frac{\pi}{4} - \frac{\pi}{2}\right)$$

[Phase shift]

$$\therefore y_1(t) = 8 \cos\left(\omega t - \frac{\pi}{4}\right)$$

Recall phasor representation.

$$\tilde{x}(t) = \tilde{x}_m e^{i\omega t}$$

$$\tilde{x}_m = x_m e^{i\phi}$$

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$$\tilde{y}_1(t) = \tilde{y}_{1,m} e^{i\omega t}$$

$$= 8 e^{-i\pi/4} e^{i\omega t}$$

Solution (con't)

Using Euler's formula: $e^{i\theta} = \cos\theta + i\sin\theta$

$$\tilde{y}_{1,m} = 8e^{-\frac{i\pi}{4}} = 8\cos\left(\frac{-\pi}{4}\right) + i\,8\sin\left(\frac{-\pi}{4}\right)$$

$$= 4\sqrt{2} - i\,4\sqrt{2}$$

Similarly for $y_2$:

$$y_2(t) = 6\sin\left(\omega t - \pi/6\right)$$

$$= 6\cos\left(\omega t - \frac{\pi}{6} - \frac{\pi}{2}\right)$$

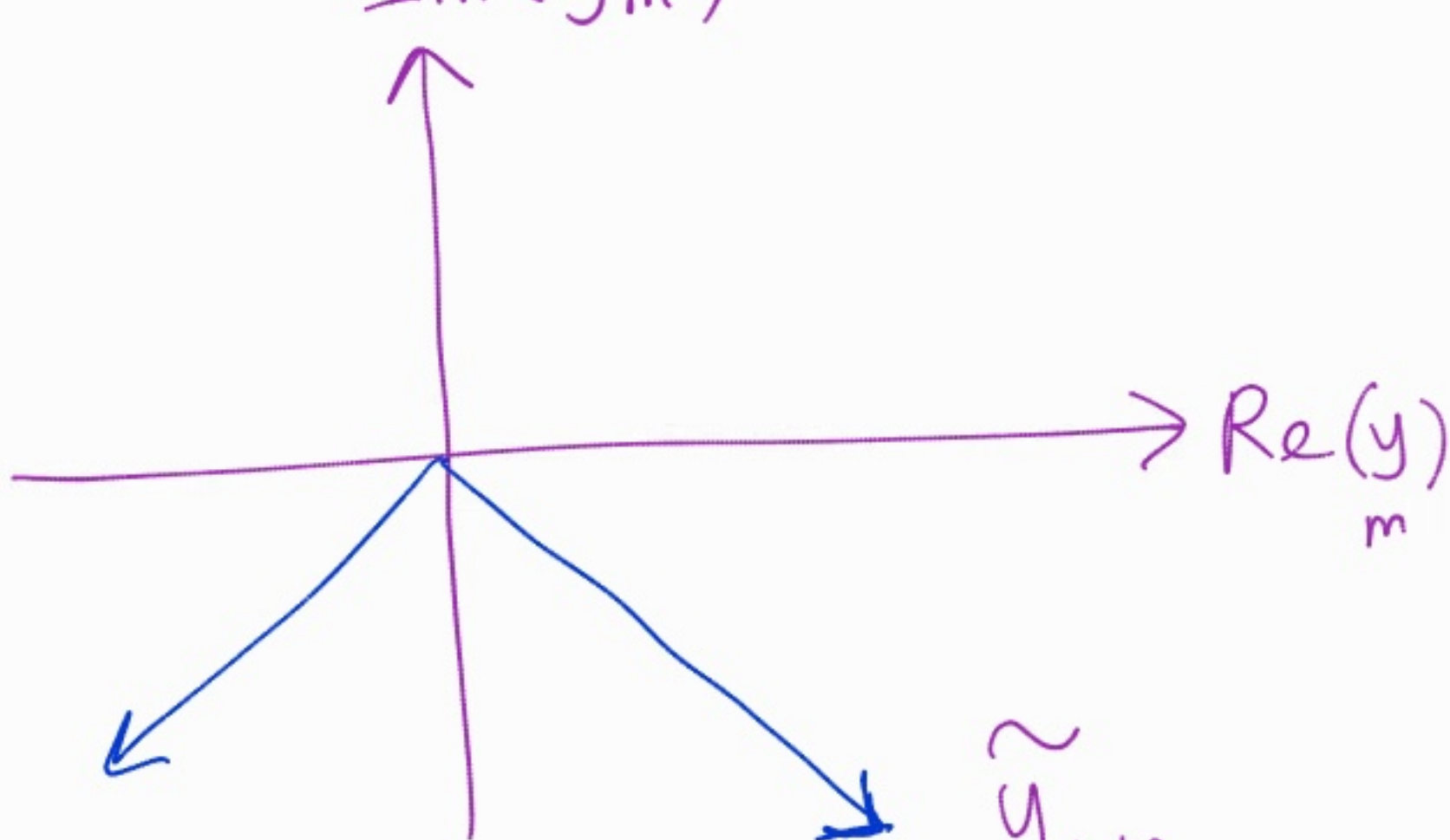
$$= 6\cos\left(\omega t - \frac{2\pi}{3}\right)$$

## Solution (cont'd)

$$\tilde{y}_{2,m} = 6e^{-i2\pi/3}$$

$$= 6\left[\cos\left(\frac{-2\pi}{3}\right) + i\sin\left(\frac{-2\pi}{3}\right)\right]$$

$$= -3 - i3\sqrt{3}$$




$\tilde{y}_{2,m} = -3 - i3\sqrt{3}$

$\tilde{y}_{1,m}$
$= 4\sqrt{2} - i4\sqrt{2}$

b) Determine the amplitude & phase of the resultant oscillation.

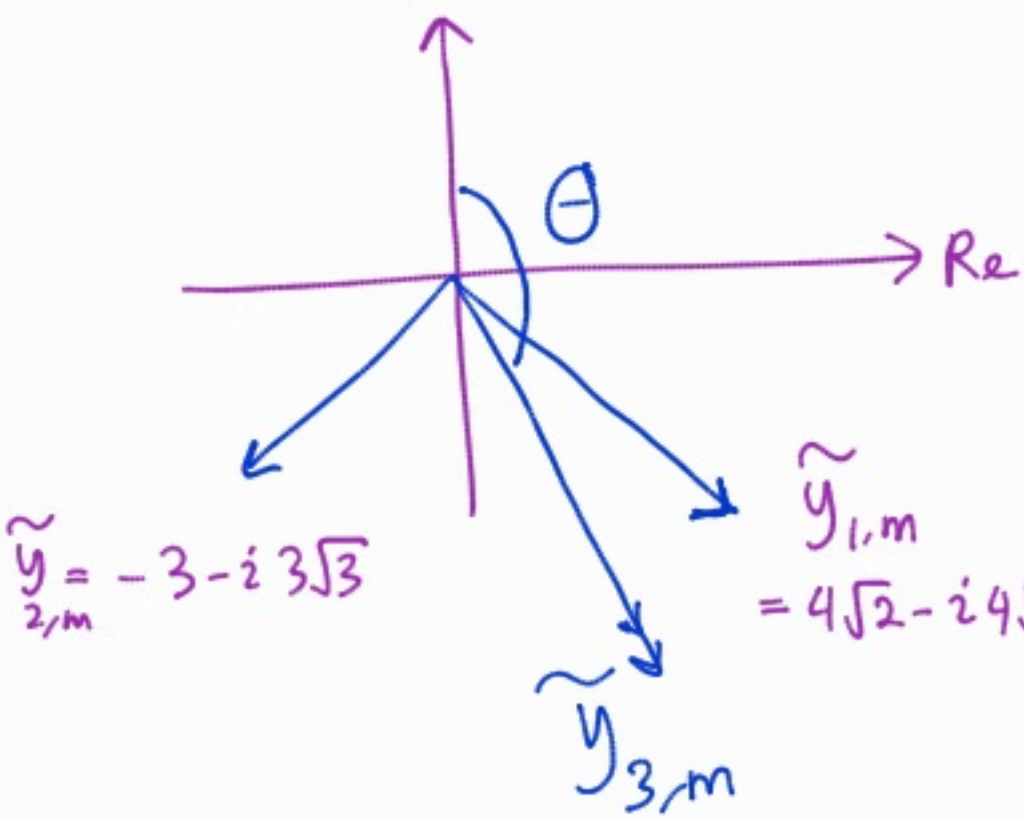
Solution.

Resultant amplitude:

$$\tilde{y}_{3,m} = \tilde{y}_{1,m} + \tilde{y}_{2,m}$$

$$= 4\sqrt{2} - i4\sqrt{2} - 3 - i3\sqrt{3}$$

$$\approx \underline{2.657 - 10.853i}$$




$$\theta = \tan^{-1}\left[\frac{-10.853}{2.657}\right]$$

$$= \underline{-76.24^\circ}$$